

IEORE 4004 - Quiz 1

This exam is composed of 12 pages and contains 2 exercises, plus a bonus exercise. The score for the bonus exercise will be computed separately and added as a bonus after the grades have been curved.

When solving an exercise, you can assume as known basic linear algebra facts, and all the statements seen in class and in the assignments, as well as closely related statements from the book and lecture notes. You cannot build on other results.

You cannot talk to your fellow students, for any reason. If you have a doubt, write your doubt in your solution, and proceed according to your interpretation of the text. You are allowed to use one regular-sized cheatsheet, handwritten by yourself. Other than that, you are not allowed to use internet, books, notes,...

Motivate appropriately all your answers. There are many empty pages to use to write your solution, or as scratch paper. If you use all of them, you can request additional scratch paper. Submit everything, including the scratch paper and the cheatsheet.

Name: _____

UNI: _____

Good luck!

(Space for the solution to Exercise 1)

1. [50 points] Consider the following Linear Program.

$$\begin{array}{ll}\text{maximize} & x_1 + 7x_2 \\ \text{subject to} & x_1 + 2x_2 = 1, \\ & x_1 - x_2 \leq -1, \\ & x_1 \text{ free} \\ & x_2 \geq 0.\end{array}$$

1. Write its dual.
2. Write the complementarity slackness conditions for this problem.
3. Using complementary slackness, decide whether the following solution is optimal:

$$(\bar{x}_1, \bar{x}_2) = \left(-\frac{1}{3}, \frac{2}{3}\right).$$

(Space for the solution to Exercise 1 and Exercise 2)

2. [50 points] Your clothing company owns one warehouse, three shops—call them A, B, and C—and sells three products: trousers, skirts, and pullovers. Your current stock at the warehouse is:

Trousers: 100 units;
 Skirts: 130 units;
 Pullovers: 200 units.

The marketing department of your company estimates the following level of fashionability of your products

Trousers: level 2;
 Skirts: level 4;
 Pullovers: level 3.

We want the total level of fashionability of the items stored at each shop to be AT LEAST the following

Shop A: 100; Shop B: 120; Shop C: 120.

The profit you make by selling a unit of these products is:

Trousers: 6\$; Skirts: 3\$; Pullovers: 8\$.

1. Assuming that all products stored in your shops will be sold and there is no cost for shipping products from the warehouse to the shops, formulate an LP that decides the quantity of each product that will be shipped from the warehouse to each of your shops so as to satisfy the constraints above and to maximize your profit (ignore the integrality constraints on the variables).
2. Now assume that each store has the following forecasted demand for each of the items

	Trousers	Skirts	Pullovers
A	40	40	40
B	10	60	40
C	20	20	70

and that shipping a unit from the warehouse to shop A, B, C has a cost of 10, 5, 6, respectively. Modify the LP above so that you find the solution that satisfies the demand in each shop while maximizing the profit minus the shipping cost.

3. To keep all our shop managers happy, we would like the total level of fashionability of each shop to be at least 80% of the level of fashionability of every other shop. (For instance, a solution where shop A has a total level of fashionability of 150 and shop B has a total level of fashionability of 200 is not allowed). Change your LP appropriately.

(Space for the solution to Exercise 2)

(Space for the solution to Exercise 2)

Bonus question. Consider the linear program

$$\begin{aligned} \max \quad & c_1 x_1 + c_2 x_2 \\ \text{subject to} \quad & Ax \leq b \end{aligned}$$

for some numbers c_1, c_2 and matrix A , and let x^* be its optimal solution. Suppose we know that:

- if c_1 is perturbed to $\hat{c}_1$ while c_2 stays constant, then x^* stays optimal as long as $\hat{c}_1 \in C_1$;
- if c_2 is perturbed to $\hat{c}_2$ while c_1 stays constant, then x^* stays optimal as long as $\hat{c}_2 \in C_2$.

We want to understand what happens if we perturb c_1 and c_2 simultaneously. In particular, let C be the set of all cost vectors $(\hat{c}_1, \hat{c}_2)$ such that x^* is optimal. Decide which of the following statements is true, thoroughly motivating your answer as appropriate.

- a) For all instances and all $(\hat{c}_1, \hat{c}_2) \in C$, we have that $\hat{c}_1 \in C_1$ (In other words, the set of perturbations of c_1 that keep x^* optimal shrinks when both c_1 and c_2 are perturbed).
- b) For all instances and all $\hat{c}_1 \in C_1$, there exists $\hat{c}_2$ such that $(\hat{c}_1, \hat{c}_2) \in C$ (In other words, the set of perturbations of c_1 that keep x^* optimal does not shrink when both c_1 and c_2 are perturbed).
- c) There are instances where both a) and b) are false.

(Extra Space for solutions to the exercises)

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